

IT Support Project Portfolio

IT Help Desk Project Documentation

Project Title: User Support & System Troubleshooting Simulation

Objective

To simulate real-world help desk scenarios by diagnosing, troubleshooting, and resolving common IT support issues in a Windows environment while applying basic cybersecurity principles.

Environment Setup

Operating System: Windows 10

Devices: Personal Laptop + Secondary Test User Account

Network: Home Wi-Fi Router

Tools Used:

Command Prompt (ipconfig, ping, tracert)

Remote Desktop Connection

Windows Defender Antivirus

Scenario 1: Network Connectivity Issue

Problem:

User unable to access the internet despite being connected to Wi-Fi.

Steps Taken:

Verified physical connection and Wi-Fi status

Checked IP configuration using ipconfig

Identified invalid IP address (169.254.x.x)

Renewed IP using ipconfig /release and ipconfig /renew

Tested connectivity using ping [google.com](https://www.google.com)

Restarted router to resolve DHCP issue

Resolution:

Internet connectivity restored after renewing IP and resetting router.

Key Skills Demonstrated:

Network troubleshooting

Understanding of DHCP and IP addressing

Logical problem-solving

Scenario 2: Slow System Performance

Problem:

User reports system is very slow during startup and usage.

Steps Taken:

- Checked Task Manager for high resource usage
- Identified unnecessary startup programs
- Disabled non-essential startup applications
- Performed full antivirus scan using Windows Defender
- Removed detected malware threats

Resolution:

System performance improved significantly after optimization and malware removal.

Key Skills Demonstrated:

- Performance troubleshooting
- Malware detection and removal
- System optimization

Scenario 3: User Account Access Issue**Problem:**

User unable to log into system due to incorrect password.

Steps Taken:

- Verified user identity
- Accessed administrator account
- Reset user password via Control Panel
- Advised user on strong password practices

Resolution:

User regained access to system successfully.

Key Skills Demonstrated:

- User account management
- Basic security practices
- Customer support communication

Scenario 4: Software Installation Issue**Problem:**

User unable to install application due to insufficient permissions.

Steps Taken:

- Reviewed error message
- Identified lack of administrative privileges
- Re-ran installer as Administrator
- Verified successful installation

Resolution:

Application installed and functioning correctly.

Key Skills Demonstrated:

- Software troubleshooting
- Permission management

Windows OS knowledge

Security Considerations

- Applied antivirus protection using Windows Defender
- Educated users on phishing and safe browsing
- Enforced strong password policies

Scenario 5: Simulated Phishing Awareness Project

Tools Used: Microsoft Outlook, Microsoft Defender, Google Workspace, Canva, Kali Linux (lab only), GoPhish (simulation platform)

Overview

Designed and conducted a simulated phishing awareness exercise in a controlled lab environment to understand common social engineering tactics and improve email security awareness. The project focused on identifying phishing indicators, analyzing suspicious emails, and documenting best practices for end-user protection.

Responsibilities

- Created simulated phishing email templates for cybersecurity awareness training
- Identified phishing indicators such as spoofed domains, suspicious links, urgent language, and fake login requests
- Configured a safe phishing simulation environment using GoPhish in a virtual lab
- Analyzed email headers and sender authentication methods (SPF, DKIM, DMARC)
- Documented incident response steps for handling phishing attempts
- Educated users on recognizing and reporting suspicious emails
- Generated a security awareness report with recommendations for reducing phishing risks

Key Skills Demonstrated

- Help Desk Troubleshooting
- Email Security Awareness
- Social Engineering Analysis
- Incident Documentation
- Microsoft 365 Security Basics
- User Support & Communication
- Cybersecurity Fundamentals

Outcome

Successfully demonstrated the ability to recognize, analyze, and document phishing threats while

applying security best practices in a safe and ethical environment.